Goals and vision

My overarching goal is to understand the computations supporting **natural behavior**. Although such behavior appears complex, they can be *decomposed* into simpler components that be tractably studied through laboratory tasks (Fig. 1A). Conversely, studying laboratory tasks under a *unified framework* – instead of in isolation – allow us to gain insight about core computations underlying natural behavior.

Instead of simply describing what the brain does in each task, my research uses **normative models** [S20] to ask what it *should* do. The three fundamental problems the brain faces during natural behavior are:

1. How to choose actions while only accessing the world through impoverished observations?

Partially Observable Markov Decision Processes (**POMDPs**) [1] provide the normative solution by positing that the brain must choose actions that maximize utility (e.g. total reward [2]) by learning and using an 'internal world model'. This is performed through three linked computations (Fig. 1B):

- **Perception**: Performing Bayesian inference to get the beliefs over latent variables in the 'world model' by combining past information (prior) with information from current observations (likelihood) [3, S1].
- **Planning**: using those beliefs to select actions that maximize expected future reward, formalized through reinforcement learning [4, S13].
- **Learning**: Use the feedback after choosing actions to update the parameters of the 'world model' – through Bayesian inference – but on longer timescales [5, 6, S8].

2. How should an agent behave when the environment or goals change?

Contextual inference (Fig. 1C) extends the POMDP framework by allowing an agent to maintain multiple candidate contexts, i.e. 'world models' [11, 12, S5, S6] , infer which context is currently active, and use the corresponding POMDP to generate actions–supporting flexible behavior in dynamic environments [7–10].

3. How should an agent act when normative solutions are computationally expensive?

Resource-rationality (Fig. 1D) extends the normative models by positing that agents balance performance with computational, energetic, and representational costs [13, 15, 14, 16, 17, S3, S17, S19] .

These three problems–and their corresponding solutions–form the **Resource-rational Context-dependent POMDP (RCP)** framework, a unifying account of how perception, learning, and planning in a **POMDP** interact with contextual inference and resource-rational constraints to support natural behavior in uncertain and changing environments.

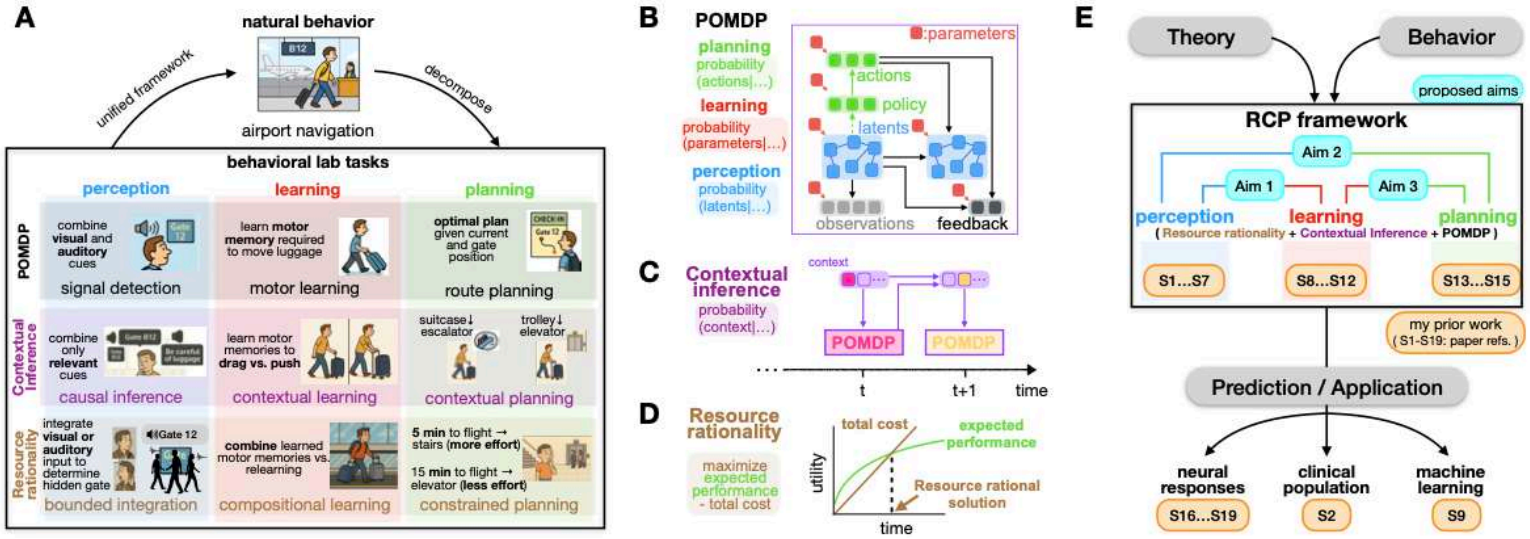

Figure 1: (A) Natural behavior–illustrated with airport navigation–arises from the interaction of three core computations: **perception**, **learning**, and **planning**. These map onto three normative principles: **(B) POMDPs** for optimal action under uncertainty,**(C) contextual inference** for identifying the active world model, **(D)** and **resource rationality** for balancing utility with computational cost. Together, this 3×3 structure offers a unified framework to study natural behavior. **(E)** Overview of the BRAIN-lab research program, which uses the RCP framework to build on my previous work studying perception, learning, and planning in isolation (orange boxes, described in publication list). Over the next five years, Aims 1–3 will combine theory and behavioral experiments to reveal how these components interact and to generate neural, clinical, and machine-learning predictions in collaboration with partner labs.

The main goal of the Behavior as Resource-rational Adaptive INference lab (**BRAIN-lab**) is to study perception, learning, and planning–traditionally examined in isolation – within this unified RCP framework (Fig. 1E). It will do so by combining formal theoretical models with behavioral data using carefully designed tasks that reveal how these computations operate. The fitted models will aid prediction/application, generating testable neural signatures[S16, S18], clinical markers[S2], and ML insights [S9]. Building on my prior work across these domains (orange boxes in Fig. 1B, described in publication list), the five-year plan systematically probes each pairwise interaction (perception & learning, perception & planning, learning & planning – cyan boxes in Fig. 1B), providing a structured path toward a long-term integrated account of how these processes jointly support natural behavior in complex, changing environments.

Five-Year Research Plan

My five-year research plan builds on the Resource-rational Context-dependent POMDP (RCP) framework to investigate how perception, learning, and planning interact to produce natural behavior. Having previously examined each process in isolation, I will now study their pairwise interactions before integrating them into a unified computational architecture:

Aim 1: Interaction between perception and learning

Natural scenes contain many objects that differ in shape, color, position, and depth, and their physical relations generate the cues we observe (Fig. 2A). Occlusion, ITD-based auditory delays, and motion or lifting forces provide information about depth ordering, support, and grouping (Fig. 2B–C).

These cues often conflict: echoes can reverse ITD–occlusion relations, and motion cues can contradict physical support (e.g., smooth objects sliding freely), producing competing 'match vs. mismatch' interpretations (Fig. 2D). A common example is an airport glass partition, where an image may be either a reflection or a person behind the barrier, creating ambiguous depth.

Illusory stimuli [18–20] exploit such ambiguities, yielding distinct percepts–changes in rotation, color, or motion direction–depending on the inferred structure (Fig. 2E). Aim 1 leverages these controlled, multi-cue scenes to reveal how perception and learning jointly infer object relationships.

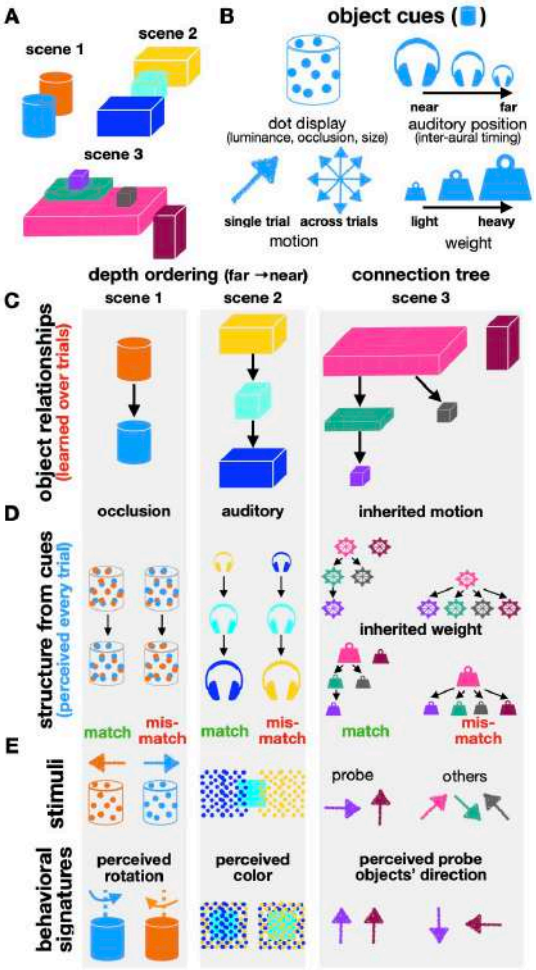

Figure 2: Interaction between perception and learning in inferring scene structure. **A.** Participants view multi-object scenes differing in shape, color, and position. **B.** Objects provide depth and relational cues–luminance, size, occlusion, ITD, motion linkage, lifting/weight. **C.** These cues jointly specify hidden structure: depth ordering and which objects rest on or move with others. **D.** Cues may agree or conflict: occlusion or ITD can reverse inferred depth (scenes 1–2), and motion or lifting cues can support or contradict attachment (scene 3). **E.** Structure-dependent illusions reveal these inferences: perceived rotation, cube color, and probe-object motion reflect the inferred depth and connection tree.

Aim 1.1: How do newly learned object relationships guide perception?

Perceptual illusions reflect structural priors shaped by natural scene statistics [21, 22]. In the RCP framework, these priors arise from long-term learning [S7], suggesting that exposure to novel structures should create new statistical expectations [23]. Aim 1.1 tests whether such learned structure directly shapes perception.

Participants first learn probabilistic cue relations in multi-object scenes, with each trial isolating one cue type (luminance, size, occlusion, ITD, motion linkage, lifting/weight) under passive viewing or structure-based judgments, allowing us to measure how quickly structural expectations form.

After learning, participants view ambiguous, structure-dependent illusions (rotating cylinder, color–depth, Johansson [24]). If cue–structure mappings are updated, percepts should shift toward the learned interpretation. Auditory or physical cue training (ITD, lifting) with transfer to visual illusions tests cue-specificity vs. cross-modal generality. RCP model fits will track how updated priors generate emergent perceptual biases.

Aim 1.2: How does the inferred perceptual structure guide learning?

Aim 1.2 asks the converse of Aim 1.1: when cues differ in consistency, how does perceptual organization influence learning of statistical object relations? Participants will view multi-cue objects whose visual, auditory, and dynamic features (occlusion, motion linkage, ITD, lifting/weight) are either consistent or conflicting, and must infer depth, attachment, and grouping on each trial.

We hypothesize that consistent cues drive observers toward a single structural hypothesis, supporting faster and more generalizable learning, whereas conflicting cues promote multiple competing hypotheses, slowing learning and increasing variability.

This aim extends cue-combination work from estimating a single latent variable to inferring full relational structure. We will regularly probe the evolving organization using explicit reports [25] and implicit sensitivity measures [S1].

Aim 1.3: How do inferred relations scale with complexity?

As scenes become richer–with more objects, relations, and dependencies–perceptual organization and structural learning grow more challenging. Aim 1.3 examines how the brain approximates inference under increasing complexity.

We will vary scene complexity along three axes: (a) no. of consistent vs. conflicting features (shape, motion, occlusion, ITD, lifting), (b) no. of participating objects, and (c) relational order from pairwise to higher-order dependencies. Each trial measures the inferred perceptual structure (Aim 1.1) and learning of latent relations across repetitions (Aim 1.2).

We hypothesize that denser or more hierarchical graphs drive observers toward compositional [S8] or heuristic approximations rather than full Bayesian inference. Consistent relations should permit modular learning, whereas conflicting or higher-order ones force broader integration, slowing learning and increasing uncertainty. Jointly tracking percepts and evolving relational models reveals how the brain manages structural complexity.

Aim 2: Interaction between perception and planning

Perception and planning are tightly linked: perception provides the beliefs on which actions rely, and planned actions can refine those beliefs. For example, retrieving a partly occluded bag requires inferring its position from uncertain cues while using movement to reduce that uncertainty. Actions thus serve both goal achievement and information gathering.

To study this interaction, participants perform a reaching task [26] in which they guide a cursor to a target (Fig. 3A) using a robotic manipulandum, eye movements, or keypad inputs. Aims 2.1–2.3 adapt this task so that the target or cursor is hidden, forcing actions to be chosen under state uncertainty. This extends optimal-feedback-control paradigms [27] to settings where cursor and target positions must themselves be inferred [28].

Aim 2.1: How does perceptual evidence integration guide planning?

Aim 2.1 examines how perception guides action when both target and cursor are hidden and must be estimated from noisy, time-extended evidence. For example, when waiting for a bag on a crowded carousel, we integrate evolving cues before acting.

Participants perform a reaching task in which cursor and target positions are inferred from visual evidence. In one paradigm (Fig. 3B, left), random-dot clouds are continuously resampled, with their mean indicating target and cursor locations. In another (Fig. 3B, right), moving blue/yellow dots [S4] give target direction (motion) and distance (yellow–blue ratio). In both, participants integrate uncertain information over time before reaching. Manipulating cue reliability controls the level of perceptual uncertainty.

Aim 2.2: How does planning adapt when actions reduce perceptual uncertainty?

Participants perform a reaching task in which cursor and target remain hidden, but movement reduces different sources of uncertainty [28, 29]. On each trial, cursor uncertainty, target uncertainty, or both may be high (Fig. 3C), indicated by diffuse dot clouds. Moving right reduces cursor uncertainty, whereas moving up reduces target uncertainty. Thus, effective behavior requires information-seeking actions: participants may first move to sharpen state estimates before reaching. By varying which uncertainty dominates, we can quantify how participants trade off uncertainty reduction against goal-directed action.

Aim 2.3: How do perception and planning interact to perform contextual inference?

Aims 2.1 and 2.2 show that perception guides planning and vice versa, but real-world behavior often requires both to infer the active context–for example, identifying which moving baggage carousel is ours. Aim 2.3 tests this joint contextual inference.

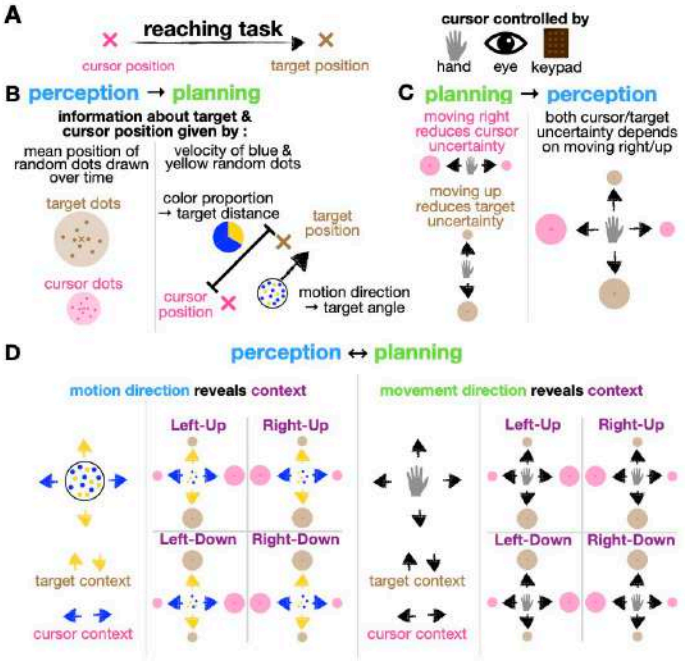

Figure 3: Interaction between perception and planning in reaching under uncertainty. **A.** Participants perform a reaching task in which a cursor (pink) is controlled by hand, eyes, or keypad to reach a target (brown). **B.** Cursor and target may be hidden. In one paradigm (left), random-dot clouds reveal their locations via time-averaged means; in another (right), dot motion gives target direction and the yellow–blue ratio encodes distance via a radial color gradient. **C.** Movements reduce perceptual uncertainty. Trials manipulate whether cursor, target, or both uncertainties are high. Moving right reduces cursor uncertainty; moving up reduces target uncertainty, requiring information-seeking trajectories. **D.** In a multi-context extension (Right/Left × Up/Down), movement–uncertainty contingencies flip. Context is inferred from motion cues (yellow = target context, blue = cursor context) or from how uncertainty changes after movement, requiring joint inference over state, context, and action.

We extend the reaching task to a multi-context setting where the mapping between movement direction and uncertainty reduction flips across four contexts (Right/Left × Up/Down; Fig. 3D). Moving right may increase or decrease cursor uncertainty depending on context, and likewise for moving up and target uncertainty. Participants must infer the active contingency from (i) motion cues indicating a cursor- vs. target-context, and (ii) how their movements change uncertainty. Sensory- and action-driven reductions jointly reveal the latent context.

Participants therefore infer the hidden state, the context, and the value of actions that shape future information. By manipulating cue reliability and movement-dependent uncertainty changes, Aim 2.3 tests the RCP prediction that perception and planning form a unified inference process, extending Kalman filtering from state estimation to context estimation.

Aim 3: Interaction between learning and planning

Learning and planning are tightly linked: learning builds the internal models enabling effective action, and planning shape the future that drive further learning. For example, adapting movements across different surfaces going through an airport.

In the RCP framework, this coupling appears as two linked inferences on different timescales: learning updates the generative model used to choose actions, and planning selects actions that reduce model uncertainty.Aim 3 probes this bidirectional interaction in structured, multi-context environments with region-specific dynamics (Fig. 4).

Aim 3.1: How are learned memories composed to support flexible planning?

Aim 3.1 asks how planning uses learned, context-specific dynamics when movements span multiple local motor contexts.

Participants first learn tool–dynamics associations (Fig. 4A), with each tool paired to a distinct curl force field [S12]. Short, local reaches bind each internal model to the tool cue, yielding multiple tool-specific forward models. They then perform long reaches across tiled regions (Fig. 4B), requiring compositional planning [30]: integrating local models into a global plan and anticipating changes at region boundaries. By varying which regions lie along the path, we test whether participants (a) relearn actions, (b) plan as if forces are absent and retrieve memories reactively, or (c) appropriately compose the learned models.

Aim 3.2: How does active exploration accelerate learning of structure?

Aim 3.2 examines the reverse of Aim 3.1: how planning shapes learning by allowing participants to choose movements

that reduce uncertainty about the underlying dynamics. Participants explore the same tiled workspace, where each region has a distinct force field linked to a tool cue. They now move directly in this heterogeneous environment under active exploration (self-chosen trajectories) or passive reaching (experimenter-defined paths). Active exploration allows them to target informative regions as uncertainty evolves.

They then perform the multi-region reaches from Aim 3.1, enabling comparison of active vs. passive learning [S9, S10, S11]. By varying region layout, force-field contrast, and boundary-cue informativeness, we can distinguish structured, uncertainty-reducing exploration from undirected wandering, paralleling Aim 2.2.

Aim 3.3: How do learning and planning jointly adapt when the environment changes?

Aim 3.3 tests how planning and learning interact when a well-learned structured environment is suddenly altered [S15]. After participants acquire stable tool–dynamics associations in the heterogeneous workspace, the environment is silently modified–by permuting region layouts or changing specific force fields. The question is how they detect, localize, and relearn the altered component.

In the RCP framework, adaptation involves detecting prediction errors that signal structural change and assigning them to the correct internal model while preserving beliefs about unchanged regions. We will test for resource-rational strategies: revisiting boundaries where forces deviate from expectation, sampling trajectories that isolate the changed region, and avoiding unchanged areas. Once identified, we will assess whether participants selectively update only the altered internal model.

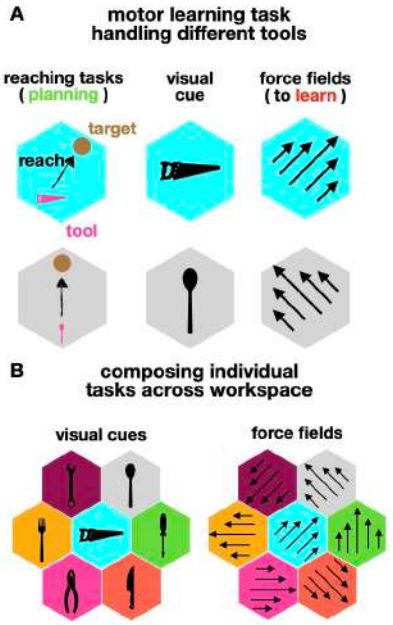

Figure 4: Interaction between learning and planning in a partitioned workspace. **A.** Participants first learn multiple tool-specific motor memories: each colored region in a hexagonal workspace requires transporting a distinct tool (pink) to a target (brown), with each tool paired with a different curl force field. **B.** After isolated learning, the tools tile the workspace, and participants make movements across regions, integrating the learned dynamics into reaches. Training directly in the heterogeneous workspace –experimenter-imposed or self-directed–tests how active curricula and experience selection shape learning efficiency.

Long-term plan: Unifying perception, learning, and planning

Aims 1–3 isolate pairwise interactions between perception, learning, and planning, laying the groundwork for a unified computational account of natural behavior. To test this fully integrated model, each aim can be extended by adding its currently missing component: **Aim 1**: Let participants act on the scene–changing viewpoints or manipulating objects–to turn passive structural learning into *active structure discovery*; **Aim 2**: Introduce gradual, within-trial changes in dynamics so participants must **update their models** online, linking perceptual uncertainty, planning, and motor learning; **Aim 3**: Add ambiguous visual or proprioceptive cues so participants infer both the governing dynamics and the **perceptual context**, integrating motor and perceptual inference.

Sustaining research

The BRAIN-Lab requires integrated behavioral and computational infrastructure. Beyond start-up support, I will pursue external funding (PM-ECRG, CRG, and others listed in Grantopedia) to establish the behavioral lab. I also welcome resource-sharing within IITGN–for example, co-using a KINARM robotic manipulandum (with Dr. Pratik Mutha) or an EyeLink eye tracker (with Prof. Meera)–to reduce costs and enhance capability.

Student Mentorship

My mentoring approach aligns with IIT Gandhinagar's interdisciplinary M.Sc. and Ph.D. programs in Cognitive Science, where students work across behavior, computation, and theory. I provide structured early guidance–training students in task design, Bayesian/RCP modeling, and data analysis–while gradually giving them autonomy to formulate and test their own ideas. M.Sc. students will take on focused, well-scoped projects such as developing behavioral tasks, running controlled experiments, or validating model components. Ph.D. students will lead full research threads within the BRAIN-Lab's three aims, owning the iterative loop between theory, modeling, and experimentation. I also plan to support co-mentorship and collaboration with IITGN's Cognitive Science faculty, ensuring students gain exposure to diverse methods and perspectives. My goal is to develop independent researchers who can navigate complex cognitive problems with quantitative rigor and creative insight.

Predictions/Applications and Potential Collaborations

Below I summarize neural, clinical, ML, and industrial applications for each aim (Fig. 1E), along with potential IIT Gandhinagar collaborators. These entries illustrate, rather than exhaust, the interdisciplinary avenues each aim naturally supports.

Aim	Neural predictions	Clinical populations	Machine learning	Industrial applications
Perception & Learning	Hierarchical relation neural codes; learning-driven prior shifts.	Atypical cue weighting; reduced prior updating.	Latent-structure ML models; probabilistic scene graphs.	Foundation model for object-relation learning from noisy cues.
Perception & Planning	Belief-state neural codes for hidden cursor/target.	Abnormal information-seeking/goal-tradeoffs.	RL benchmarks for active sensing and context-dependent control.	Active-perception modules for robots/drones.
Learning & Planning	Neural signatures of multiple forward models.	Difference in acquiring/composing dynamics.	Benchmarks for continual/curriculum learning.	Compositional motor-learning engine for robots.
IITGN collaborators	Vaibhav Tripathi, Krishna Prasad Miyapuram	Pratik Mutha, Uttama Lahiri	Krishna Prasad Miyapuram, Nipun Batra	Vineet Vashista, Yogesh Kumar Meena

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